



Flight

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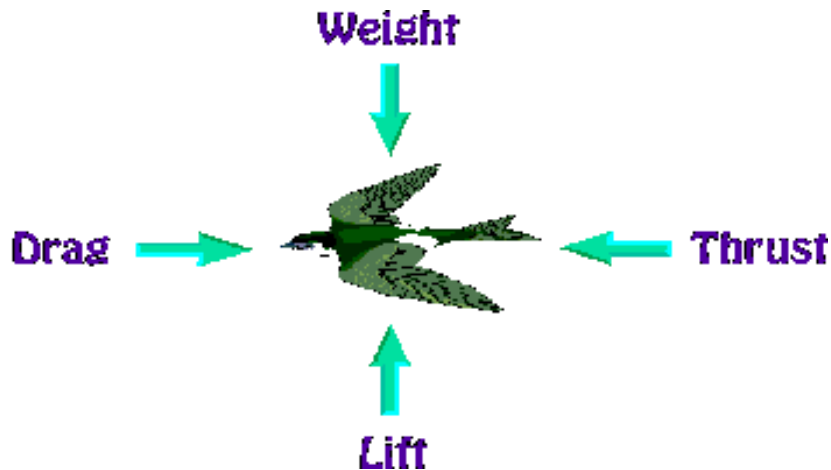
Flight!

- ▶ Energetically expensive per unit time (= burn lots of calories each minute)
- ▶ Energetically efficient per unit distance (= go farther for each calorie)



https://www.youtube.com/watch?v=ogeEY2q3_OU

Flight aerodynamics



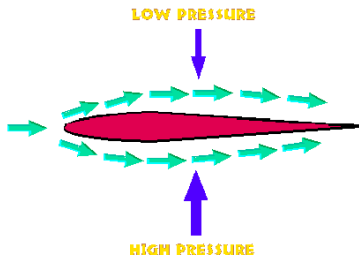
(MIT)

Flight aerodynamics: Bernoulli lift

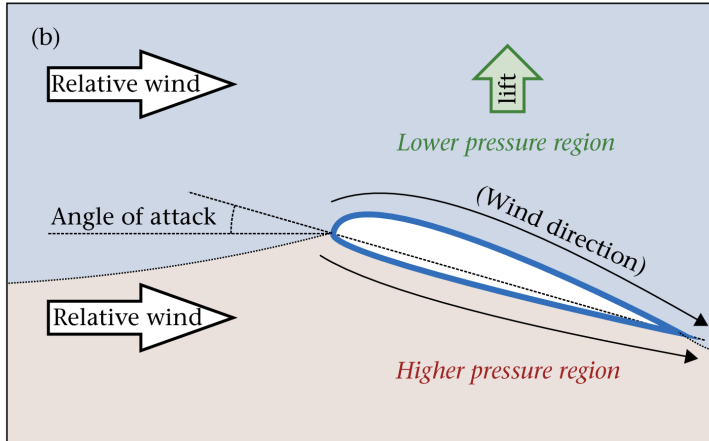
$$P_{static} + P_{dynamic} = P_{total}$$

$$\rho * A * V = C$$

(where P is pressure, ρ is density, A is area, V is velocity, and C is a constant—increase the area above a wing with constant air density, and velocity must increase, reducing static pressure and generating **lift**)



Flight aerodynamics: Bernoulli lift



(Bajec & Heppner 2009)

Which combination will increase Bernoulli lift?

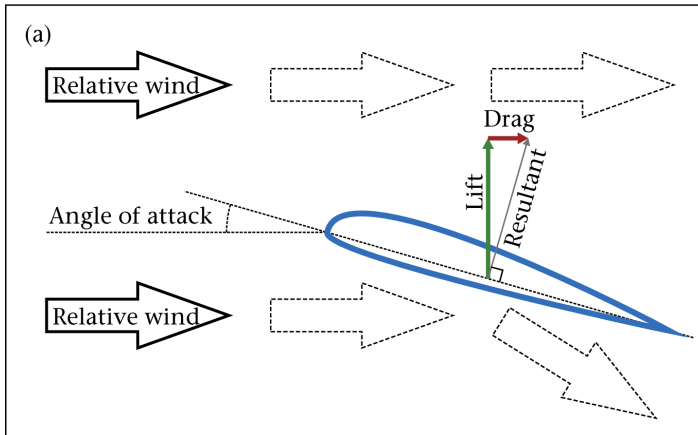
$$P_{static} + P_{dynamic} = P_{total}$$

$$\rho * A * V = C$$

- a) larger wings and constant speed
- b) smaller wings and constant speed
- c) greater speed and larger wings
- d) a & c
- e) impossible to say

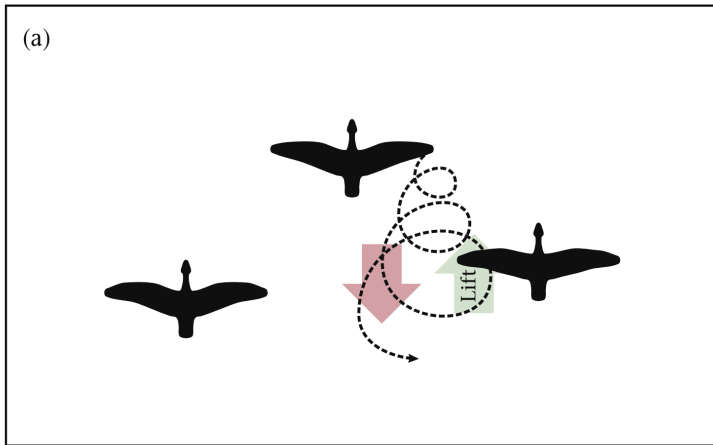
Flight aerodynamics: Newtonian lift

(“equal and opposite reaction”: air is deflected downward when there is a positive angle of attack)



Flight aerodynamics: lift from vortices

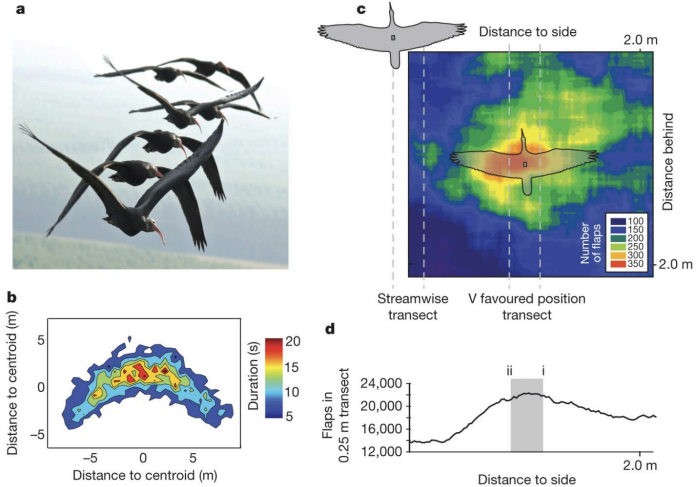
(spirals of air off a leading bird's wing can provide lift to a follower)



(Bajec & Heppner 2009)

Ibises exploit lift from vortices!

(most flaps occur in theoretically optimal position)



(Portugal et al. 2014)

Drag

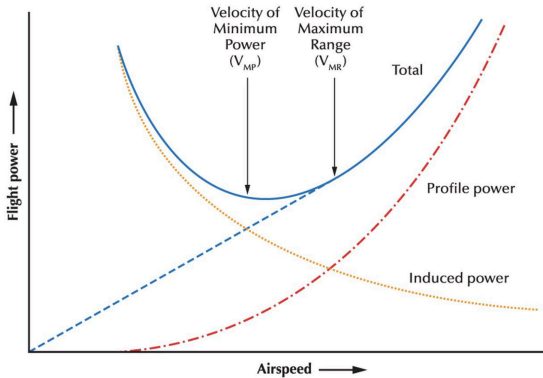
- ▶ **induced drag:** drag from air streams deflecting from wing—reduced by alula!
- ▶ **parasitic drag:** friction between bird and air
- ▶ **profile drag:** related to form of wing



(Portugal et al. 2014)

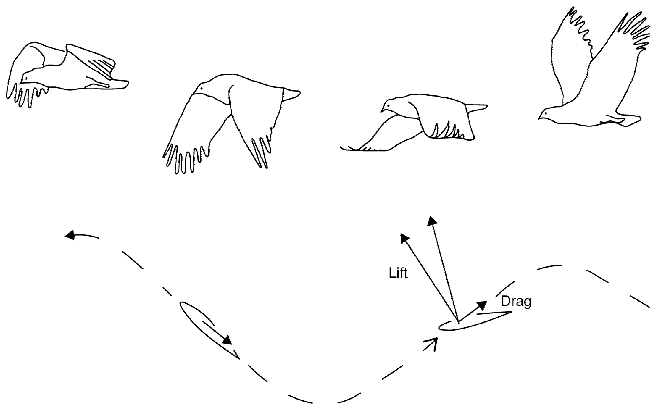
Flight energetics

- ▶ Upper left: high power needed to overcome weight
- ▶ Upper right: high power needed to overcome drag
- ▶ V_{MP} : speed with minimum energy use per unit time
- ▶ V_{MR} : speed for maximum distance per unit energy



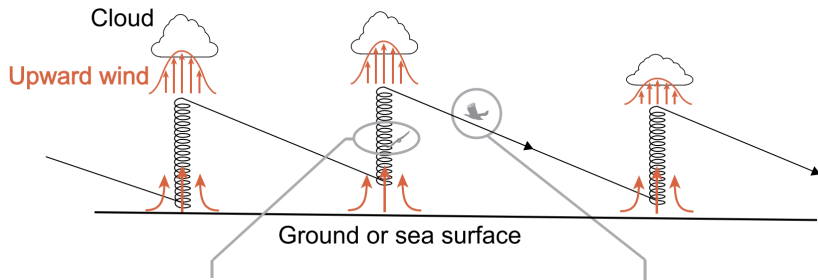
Flapping generates lift and thrust

- ▶ **Flight stroke** consists of **upstroke** and **downstroke**
- ▶ rotating wing forward generates forward thrust, rotating back minimizes drag (“flap, fold, and twist”)
- ▶ from 2x / second (Gray Heron) to ~ 80 x / second (hummingbirds)



Thermal soaring

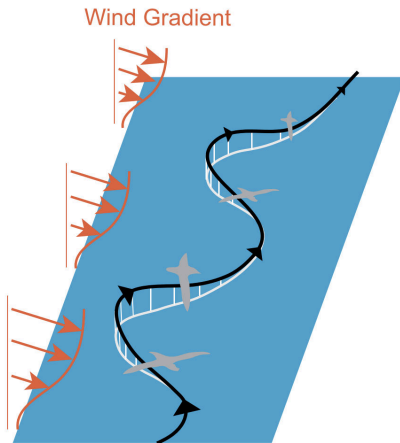
- ▶ Thermal soarers exploit rising hot air columns, gliding from one to another
- ▶ Examples: vultures, frigatebirds



(Goto et al. 2022)

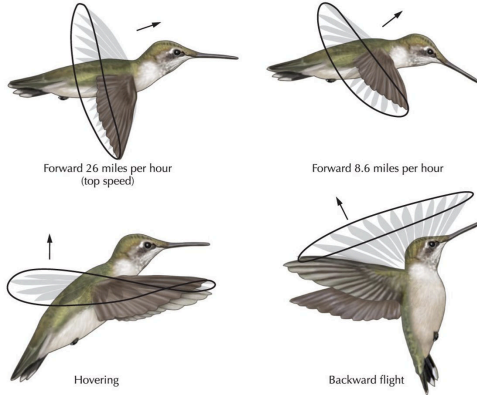
Dynamic soaring

- ▶ Dynamic soarers exploit deflected wind, turning into currents to rise and away to glide forward
- ▶ Examples: pelagic seabirds, hawks along mountains



Hummingbird hovering

- ▶ Figure-8 motion
- ▶ Up to 78 cycles (upstrokes + downstrokes) per second!



https://www.youtube.com/watch?v=gJ_T_Y1rxHw

Flight ecomorphology

- ▶ Wing aspect ratio = $\frac{\text{Wingspan}}{\text{Wing area}}$
- ▶ Wing loading = $\frac{\text{Body mass}}{\text{Wing area}}$

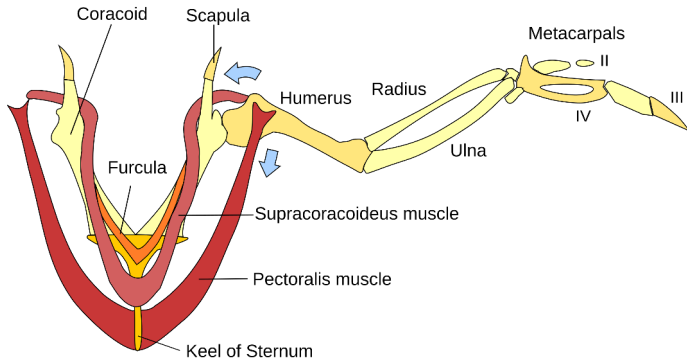


Which species would you expect to have the lowest wing aspect ratio?

- a) a frugivorous parrot
- b) a pelagic seabird
- c) a hummingbird
- d) a grouse
- e) a migratory songbird

Major wing muscles and furcula

- ▶ **pectoralis** powers the downstroke
- ▶ **supracoracoideus** operates as pulley through triosseal canal (gap between scapular, coracoid, furcula) to power upstroke
- ▶ **furcula**: expands and contracts with wing beats; may enhance respiratory performance



References

- ▶ Title slide species: *Odontophorus melanotus*, *Pyriglena leuconota* *Xiphorhynchus flavigaster*, *Pteroptochos tarnii*
- ▶ Bajec, I. L., & Heppner, F. H. (2009). Organized flight in birds. *Animal Behaviour*, 78(4), 777-789.
- ▶ Portugal, S. J., Hubel, T. Y., Fritz, J., Heese, S., Trobe, D., Voelkl, B., . . . & Usherwood, J. R. (2014). Upwash exploitation and downwash avoidance by flap phasing in ibis formation flight. *Nature*, 505(7483), 399-402.
- ▶ Linehan, T., & Mohseni, K. (2020). Scaling trends of bird's alular feathers in connection to leading-edge vortex flow over hand-wing. *Scientific reports*, 10(1), 7905.
- ▶ Gill, F. B., Prum, R. O., & Robinson, S. K. (2019). *Ornithology* (4th ed.). W. H. Freeman.

References

- ▶ Shyy, W., Aono, H., Kang, C. K., & Liu, H. (2013). An introduction to flapping wing aerodynamics (Vol. 37). Cambridge University Press.
- ▶ Goto, Y., Yoda, K., Weimerskirch, H., & Sato, K. (2022). How did extinct giant birds and pterosaurs fly? A comprehensive modeling approach to evaluate soaring performance. PNAS nexus, 1(1), pgac023.